Internship Overview

Vegetation and Obstacle Perception

[ID 012026]

Develop a perception pipeline to estimate grass height detect invasive weeds and identify field obstacles for a small ground robot.

Objectives

- Estimate grass height in field conditions
- Detect invasive weeds in camera imagery
- Identify poles and other obstacles around the robot

Tasks

- Train object detection model
- Deploy a lightweight object detection model
- Interface hardware
- Validate performance usign the robot chassis

Requirements

- Python programming
- Computer vision fundamentals (OpenCV)
- AI libraries fundamentals (Pytorch, Tensorflow)
- Linux OS fundamentals

Equipment

- Jetson Orin Nano
- Teltonika RUT240 4G LTE Router
- VL53L0X ranging sensors
- Arducam IMX219 NoIR camera
- UGV Rover ROS 2 Open-source 6 Wheels 4WD

Deliverables

- Perception prototype for grass height, weeds and obstacle detection
- Test results from field trials

Work Breakdown

- 20% hardware integration
- 80% coding

Duration: 2 months

Payment: Available in Bizquick

Supervisor: Kaleb Granados